

RUSHI ASHOK SAVANI

+1 (315) 952 4519 | rushiashokbhaisavani@gmail.com | [Linkedin](#) | [Github](#) | [Portfolio](#)

SUMMARY

Applied AI/ML engineer specializing in production AI systems, with experience spanning predictive modeling, large language model workflows, and cloud-native deployment. Proven ability to take models from development through validation to production release in regulated, compliance-sensitive environments. Seeking full-lifecycle ownership of ML infrastructure where engineering rigor directly drives operational and financial outcomes.

WORK EXPERIENCE

AI/ML Engineering Intern | *ProHealth Group, Atlanta GA*

Jun 2025 – May 2026

- Architected a production RAG system using AWS Bedrock, Titan embeddings, OpenSearch Serverless, Claude SQL generation, and per-user caching, enabling administrators to retrieve clinical and financial insights in under 10 seconds.
- Trained an XGBoost rehospitalization classifier on 12 clinical and claims features with SMOTE-Tomek resampling to correct a 4:1 class imbalance, lifting risk-targeting precision from 64% to 82% across a 55K patient holdout set.
- Shipped LLM-powered investigation workflows inside an internal SaaS product with cited explanation, iterative prompt engineering and structured eval plans to validate anomaly detection outputs, saving an estimated \$100K/year in manual analysis.
- Productionized ML and LLM workflows as FastAPI services with Dockerized, CI/CD-backed AWS releases, adding CloudWatch logging and Terraform-managed infrastructure to reduce manual deployment risk.

Research Assistant | *Syracuse University, School of Information Studies*

Jan 2025 – May 2025

- Fine-tuned a RoBERTa-base text classification model with HuggingFace Transformers on 5,000+ labeled AI-adoption survey responses, using tokenization and validation tracking to classify five organizational readiness segments with 91% accuracy.
- Automated survey data cleaning, statistical summaries, and dashboard-ready output generation in Power BI using Python, pandas and scipy, reducing the research team's reporting cycle from two weeks to three days.

Data Analyst | *Krishna Jewels, India*

Jun 2023 - Jun 2024

- Designed a relational MySQL data model across inventory, sales, supplier, and accounting records, digitizing 40K+ records and reducing manual lookup time by 70%.
- Surfaced slow-moving SKUs and high-demand items through SQL-based sales trend analysis, improving gross margin by 12%.
- Centralized transactional data with data visualization into Power BI dashboards with daily scheduled refreshes, giving stakeholders self-serve access and eliminating 15+ hours/week of manual reporting.

TECHNICAL SKILLS

- Programming & Backend:** Python, SQL, FastAPI, REST APIs, TypeScript
- Machine Learning & Statistics:** Scikit-learn, XGBoost, PyTorch, TensorFlow, HuggingFace Transformers, Feature Engineering, Model Evaluation, Hyperparameter Tuning, A/B testing
- GenAI & NLP:** RAG, LangChain, LangGraph, LiteLLM, Vertex AI/Gemini, HuggingFace Transformers, Vector Databases (Qdrant, AWS Titan embeddings), Prompt Engineering, Text Classification
- Cloud & MLOps:** AWS (EC2, ECS, S3, Secrets Manager), Microsoft Azure, Azure AI, Docker, GitHub Actions CI/CD
- Data Engineering & BI:** PostgreSQL, MySQL, Data modeling, Snowflake, DBT, Fivetran, ETL Pipelines, Power BI, Tableau

PROJECTS

VaultMind | *Private On-premises Q&A System*

- Deployed a zero-egress RAG system for document Q&A on quantized Llama 3.1 with nomic-embed-text embeddings and Qdrant vector search, serving fully offline inference through FastAPI to meet healthcare data privacy requirements.
- Optimized retrieval accuracy for long clinical documents by layering BM25 sparse matching with dense vector search via Reciprocal Rank Fusion and reranking top candidates through a cross-encoder, cutting p50 latency by 68% on a 3B quantized local model and eliminating ~\$140/month in cloud LLM costs.

MyFinancial Advisor | *Investment Recommendation System*

- Forecasted short-term stock direction using an LSTM network on engineered price and market-signal features, outperforming a logistic-regression baseline by 14% in directional accuracy.
- Indexed 500+ financial filings and news articles as vector embeddings in Pinecone, enabling source-backed validation of investment signals through a RAG retrieval layer.

KarmaScore Predictor | *Subscription Churn Analysis*

- Predicted subscriber churn across 10K+ records using XGBoost, reducing 60 raw features to 23 through importance ranking and correlation filtering to achieve 89% precision on the churn class.
- Mapped SHAP scores into a customer-level intervention likelihood metric ranking which at-risk subscribers are worth retaining based on projected profitability, displayed in an interactive Streamlit dashboard.

Annual Report Analysis with Advanced Language Models: A Stock Investment Strategy Enhancement

- Processed data from 50+ organizational annual reports and deployed OpenAI's ChatGPT-3.5 Turbo for generating structured responses to financial analyst queries on investment trends and report interpretation and published findings in (IRJET)

EDUCATION

Master of Science | Information Systems, Data Science
Syracuse University, School of Information Studies, Syracuse, NY

May 2026
(GPA: 3.9/4)

Bachelor of Engineering | Information Technology
Thadomal Shahani Engineering College, India

May 2024
(GPA: 8.52/10)